

Periphrastic Negation in Kalaallisut

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1 Introduction

Kalaallisut (ISO 639: kal, Inuit, Greenland) has two ways of forming negation: a negative suffix (1) and a periphrastic negation construction (2), which we refer to as the PNC.^{1,2,3}

- (1) Mersu-**nngi**-langa.

sew-NEG-1SG.IND

I don't sew.

Ellen Thrane CRS-1

- (2) Mersor-**neq** **ajor**-punga.

sew-NEQ NEG.HAB-1SG.IND

I don't sew.

Ellen Thrane CSR-6

Suffixal negation is the general purpose negation strategy, whereas the PNC expresses habitual negation (section 3.2). The suffixal strategy is unremarkable given the polysynthetic nature of the language (Fortescue 1999). The periphrastic construction is more surprising and our focus here. We develop an analysis in which *ajor-* is a negative auxiliary and the *-neq* suffix a non-finite tense

¹Elicited and volunteered examples are annotated with the name of the speaker and an alpha-numerical code identifying it in the Line Mikkelsen's notebooks. Textual examples primarily come from a collection of Kalaallisut texts transcribed and glossed by Maria Bittner (<https://sites.google.com/view/maria-bittner/kalaallisut>). These are annotated with the name of the story, paragraph and line number. Examples drawn from other online resources are sourced with a URL. Examples from Bittner's text collection are rendered in standard orthography and reglossed in accordance with footnote 2.

²Glossing conventions: 1 = first person, 2 = second person, 3 = third person, 4 = fourth person (non-local person DPs that are co-referential with a higher subject are fourth person), ABL = ablative case; ABS = absolutive case; ALL = allative case; AP = antipassive; APPL = applicative; CAU = causative mood; CAUS = causativizer; COND = conditional mood; CONJ = conjunction; CONT = contemporative mood; COP = copula; DEF = definite; DEM = demonstrative; DIST = distal; ERG = ergative case; FOC = focus; FRUST = frustrative; FUT = future; HAB = habitual; IND = indicative mood; INF = infinitive; INST = instrumental; INTRANS.PART = intransitive participle; LOC = locative case; NEG = negative; NOM = nominalizer; PART = participial mood; PASS = passive; PASS.PART = passive participle; PL = plural; PRES = present; PTCP = participle; SG = singular; VIA = vialis case. 3SG>3SG.IND means third person singular subject acting on third person singular object in indicative mood. 1SG>3SG.ERG means 1SG possessor with 3SG possessum in the ergative case. As for *-neq*, Bittner (1994) glosses it GER for 'gerund', Fortescue (1984) glosses it NOM for 'nominalizer' and Kahn and Valijärvi (2022) gloss it AP for 'abstract particle'. Since the category status of *-neq* is a central issue of the paper, simply gloss it as NEQ, and rely on the discussion in sections 4–6 to clarify its status. Note that clauses unmarked for tense can receive either a present or past interpretation, including in the PNC. Other abbreviations used are: apP = Antipassive Phrase, ApplP = Applicative Phrase, C = complementizer, CausP = Causative Phrase, CP = Complementizer Phrase, D = Determiner, DP = Determiner Phrase, Neg⁰ = Negation head, NegP = Negation Phrase, PNC = Periphrastic Negation Construction, SOV = Subject-Object-Verb order, Subj = Subject, V = Verb, VP = Verb Phrase, Voice⁰ = Voice head (active or passive); VoiceP = Voice Phrase.

³A configuration of non-finite verb and finite negative auxiliary, as in (2), is also found in other languages, prominently Uralic languages, where it is known as a 'connegative' construction. See typological discussion of negative strategies at the end of this section.

head. Evidence for this analysis comes from a systematic comparison with control structures of the type illustrated in (3), which also feature a *neq*-suffix verb followed by a finite verb.

- (3) Naja-p mersor-**neq** nuannari-vaa.
 Naja-ERG sew-NEQ love-3SG>3SG.IND
 ‘Naja loves to sew.’

Grethe Schmidt CIC-GS-1

While both constructions employ the *-neq* suffix, we argue that it instantiates different syntactic categories in the two structures. In the PNC, *-neq* is a non-finite T head. In control structures it is a D head, in line with its general nominalizing function (Fortescue 1984:44ff). We present morphological, syntactic and semantic evidence for this distinction from case marking, verbal agreement, possessive inflection, scrambling, subcategorization and theta roles. While the PNC is documented in descriptive and pedagogical work (Fortescue 1984:47–48, Kahn and Valijärvi 2022:245–247, Sadock 2003:10–11, 33 and Woodbury (1975:70–71), the present paper provides, to the best of our knowledge, the first in-depth analysis of the PNC and also the first systematic comparison with control structures.

In support of our synchronic analysis, we provide a diachronic analysis of the development of the PNC that situates it in a larger group of ‘negative verbs’. We argue that *ajor-* originated as a transitive control verb meaning ‘not unable to reach or do’ that took a nominalized DP complement, akin to *nuannari-* in (3). Two intertwined changes then occurred: the nominalizer *-neq* was reanalyzed as a verbal morpheme and *ajor-* was grammaticalized into a negative auxiliary verb. We present evidence for this analysis from proto-Inuit, from comparison with other modern day Inuit varieties, and from observations internal to the grammatical system of Kalaallisut.

In addition, our demonstration that *ajor-* is a negative auxiliary is of typological relevance. Negative auxiliaries form a rare negation strategy. In the WALS sample only 47 of 1157 languages, or 4%, use negative auxiliaries. Of those 47 languages, only 3 are of the same polysynthetic type as Kalaallisut. As noted above, Kalaallisut also has affixal negation. This puts Kalaallisut with only 21 languages, or 2% of the WALS sample, that have multiple negation strategies. Of these language with multiple negation strategies only Lavukaleve (Papuan, Solomon Islands) features a negative auxiliary as one of the negation strategies.⁴ Thus the Kalaallisut negation system is very rare and an important addition to the WALS sample.

In section 2 we provide background on our collaboration and on Kalaallisut language ecology as well as relevant linguistic features of language. Section 3 introduces the PNC by illustrating its function in language use in a variety of genres and comparing it to suffixal negation. We also discuss the homophonous lexical verb *ajor-* ‘be bad’ and argue that it is distinct from the negative auxiliary. Section 4 lays out the empirical differences between the PNC and control structures that motivate an analytic distinction between the two. In section 5 we examine a number of predictions this analytic distinction makes and show that they are borne out. This then paves the way for our structural analysis of the two constructions in section 6. Here we locate the *-neq* suffix on the clausal spine and present arguments for *ajor-* being a negative auxiliary located in Neg. In section 7 we turn to diachrony and section 8 concludes the paper.

⁴See Appendix A for details. Thanks to Anonymous 1 for checking original sources for morphological type and to Anonymous 2 for identifying Lavukaleve as the only language with a negation system comparable to Kalaallisut’s.

2 Background

This section i) provides background on the collaboration behind this paper (section 2.1), ii) lays out the linguistic ecology of Kalaallisut (section 2.2) and iii) exemplifies basic linguistic features of Kalaallisut that are relevant for the analysis of the PNC and control constructions at the heart of the paper (section 2.3).

2.1 Collaboration

This work stems from a collaboration between linguists Line Mikkelsen and Tyler Lee-Wynant and Kalaallisut speakers Grethe Schmidt and Ellen Thrane.

Grethe Schmidt’s autobiography I was born in Attu in northwestern Greenland and spoke Kalaallisut at home and in school. I lived in a number of other places in Greenland before moving to Denmark in 1980. I am a Kalaallisut-Danish-Kalaallisut translator for Danish TV-channels and also affiliated with Umiaq at Kalaallit Illuutaat. Kalaallisut is the language I am born with.

Ellen Thrane’s autobiography I was born in Upernavik in northwestern Greenland. My mother and grandmother spoke only Kalaallisut, whereas my father spoke Kalaallisut and Danish. I acquired Kalaallisut from birth and only started learning Danish in school. I moved south to Ilulissat at age 9 and to Denmark after 7th grade for educational opportunities. I volunteer as a Kalaallisut-Danish-Kalaallisut interpreter for Danish public agencies and am affiliated with Umiaq, a Greenlandic association hosted at Kalaallit Illuutaat (‘The Greenlanders’ house’) in Odense, Denmark. Kalaallisut is my mother tongue and identity.

Collaboration Line Mikkelsen and Ellen Thrane began collaborating in summer of 2022, Grethe Schmidt joined the project in spring of 2024 and Tyler Lee-Wynant in Fall of 2024. The data discussed below come from one-on-one weekly Zoom elicitation sessions, which also included discussion of linguistic questions of interest. Elicitation was conducted by Line Mikkelsen in Danish. Danish prompts and translations were subsequently translated into English by the Line Mikkelsen. All elicitation recordings are archived in the California Language Archive. With the exception of the scrambling data in (44), judgments were overall uniform, within speaker and between speakers. Key data points were additionally checked with Kalaallisut speaker Agnes Johnsen and native-speaker linguist Naja Blytmann Trondhjem.

2.2 Language ecology

Kalaallisut is a member of the Inuit-Yupik-Unangan language family, which is spoken from Eastern Russia across Alaska and Canada to Greenland. It is the sole official language of Greenland, with around 50,000 speakers in Greenland and another 10,000 in Denmark.⁵ The language is acquired by children in the home and taught in school. Denmark has been/continues to be a colonial power

⁵The number of Kalaallisut speakers in Denmark reported above comes from (<https://greenlandic.dk/om-sproget/>) and is based on reports from the “Greenlandic Houses” and linguistic studies by Ilisimatusarfik (University of Greenland). There is no official government tally of Kalaallisut speakers in Denmark.

in Greenland and Danish language (ISO 639: dan) has a big presence, especially in administration. English (ISO 639: eng) has a growing presence.

2.3 Basic linguistic features

Three linguistic aspects of Kalaallisut are of immediate relevance for our study: i) polysynthesis, ii) case and iii) verbal agreement.

Kalaallisut manifests classic features of polysynthesis. First, verbs may exhibit a high morpheme-to-word ratio, as in the textual examples in (4) and (6).⁶

- (4) Qulingilua-a-su-ssa-a-galuar-put.
 nine-COP-3SG.PART-FUT-COP-FRUST-3PL.IND
 ‘There should’ve been nine.’ *Paakujuk* (Sooty), §10, (11)
- (5) Nukappiaqqa-p unit-sin-niar-aluar-pai.
 boy-ERG.SG stop-CAUS-try-FRUST-3SG>3PL.IND
 ‘The boy tried to stop them [= the dogs].’ *Paakujuk* (Sooty), §27, (2)
- (6) ... navia-nar-tor-sior-tip-paanga.
 fear-induce-NOM-experience-CAUS-3SG>1SG.IND
 ‘... it put me in danger’ [lit. it made me experience having fear induced]
Piniartup uqaluttuaa (A hunter’s story), §5, (4)

As these examples also show, there is rich verbal inflectional morphology, specifically mood and subject and object agreement. These categories are expounded together in a single, obligatory word-final suffix. Third, Kalaallisut exhibits grammatically free argument drop and noun incorporation.⁷ This is illustrated in (7), where all DP arguments are null pronouns and the notional object of ‘consume’ is incorporated.

- (7) *pro* kaagi-sor-uit *pro* *pro* ajori-ssu-akkit.
 2SG.ABS cake-consume-2SG.COND 1SG.ERG 2SG.ABS angry-FUT-1SG>2SG.IND
 ‘If you eat cake, I’ll be angry with you.’ Ellen Thrane ANM-1

Finally, Kalaallisut basic word order is SOV, but with discourse-induced flexibility (Fortescue 1984:93ff; Fortescue 1993).

Case alignment is ergative-absolutive. In (8) the transitive subject is marked with the ergative suffix *-p*. In contrast the transitive object and intransitive subject (9) are absolutive, which is null. The absolutive/ergative case distinction is only morphologically expressed in singular and possessed nouns. We nonetheless gloss all core arguments as ergative or absolutive in accord with their syntactic function.

- (8) Naja-**p** Juuna ikior-paa.
 Naja-ERG Juuna.ABS help-3SG>3SG.IND
 ‘Naja helped Juuna.’ Ellen Thrane AFX-1

⁶A reviewer points out that the word for ‘nine’ is, at least historically, morphologically complex deriving the literal meaning of ‘not quite ten’. Taking this into account would increase the morpheme-to-word ratio in (4) even further.

⁷A distinctive feature of Inuit noun incorporation is that it is lexically restricted to a set of obligatorily incorporating verbal roots.

- (9) Juuna angerla-jaar-poq.
Juuna.ABS come.home-early-3SG.IND
'Juuna came home early.'

Ellen Thrane AFX-2

There is at most one absolutive argument per clause (Compton 2018:164). Therefore, two internal arguments of a ditransitive verb cannot both be absolutive (10). Instead, one must be realized as an oblique (11).

- (10) *Meeqqa-t poortukka-t nassip-pakka.
child-PL.ABS package-PL.ABS send-1SG>3PL.IND
Intended: 'I sent the children the packages.'

Ellen Thrane CKS-2

- (11) Meeqqa-t poortukka-nik nassip-pakka.
child-PL.ABS package-PL.INST send-1SG>3PL.IND
'I send the children the package.'

Ellen Thrane CKS-3

Verbs agree with all absolutive and ergative arguments. (12) is ungrammatical because the verb only agrees with the subject. (13) is ungrammatical because the verb only shows agreement with the object. The only grammatical realization of a formally transitive clause is as in (14) where the verb agrees with both the ergative subject and the absolutive object.

- (12) *Naja-p meeqqa-t ikior-poq
Naja-ERG child-PL.ABS help-3SG.IND
Intended: 'Naja helped the children.'

Grethe Schmidt CIE-GS-1

- (13) *Naja-p meeqqa-t ikior-put.
Naja-ERG child-PL.ABS help-3PL.IND
Intended: 'Naja helped the children.'

Grethe Schmidt CIE-GS-2

- (14) Naja-p meeqqa-t ikior-pai.
Naja-ERG child-PL.ABS help-3SG>3PL.IND
'Naja helped the children.'

Grethe Schmidt CIE-GS-3

Verbs do not agree with oblique arguments, including allative (15) and instrumental (16) arguments:

- (15) Ilin-nut allap-punga/*pakkit.
2SG-ALL write-1SG.IND/1SG>2SG.IND
'I wrote to you.'

Ellen Thrane CHT-1/CHT-2

- (16) Meeqqa-nik ikiu-i-vunga/*vakka.
child-PL.INST help-AP-1SG.IND/1SG>3PL.IND
Intended: 'I helped some children.'

Grethe Schmidt CZJ-GS-3/CZJ-GS-2

In sum, verbs agree with all and only absolutive and ergative arguments.

3 Introducing the Periphrastic Negation Construction

In this section we introduce the PNC in a general way as background for the more targeted description and analysis in the following sections. We start by illustrating the use of the PNC (section 3.1). We then compare it to affixal negation in terms of semantics (section 3.2) and frequency (section 3.3). Finally we discuss the relationship between the negative auxiliary *ajor-* and the homophonous lexical verb meaning ‘be bad’ (section 3.4).

3.1 PNC usage

While the PNC is less frequent than suffixal negation (see section 3.3 below), it is robustly attested across all genres that we have access to.⁸

The example in (17) comes from *Silliarnamik uqaluttuaq* (The tale of Silliarnaq) as included in Maria Bittner’s online text collection (<https://sites.google.com/view/maria-bittner/kalaallisut>). Bittner characterizes this text as a myth. The myth follows three brothers and their friend. On a hunting trip they end up far out on the ice-covered sea. They eventually arrive in an unfamiliar land, where an old man becomes their foster father. In the translation of the sentence below ‘they’ refer to the brothers and their friend and ‘the old man’ to the foster father. The habitual meaning of the periphrastic negation is consonant with the context, specifically the use of the habitual suffix *-(g)aang* in the preceding subordinate clause (*whenever they woke up.*)

- (17) a. Ukiu-p ingerla-ner-a-ni ullaa-kkut iter-**aang**-ata
year-ERG go.on-NEQ-3SG>3SG-LOC morning-VIA wake.up-HAB-3PL.CAU
In the course of the year, whenever they woke up in the morning,
b. utoqqaar-suaq **peqar-neq ajor-poq**
old-big.SG.ABS be.there-NEQ HAB.NEG-3SG.IND
the old man was never there.

Silliarnamik uqaluttuaq (The tale of Silliarnaq), §9, (1)

The example in (18) is from a personal narrative by the Ellen Thrane. She is narrating about her childhood, including play and other activities. As in (17) the habitual component of the PNC is matched by a habitual morpheme, *-tar*, located in the following contrastive clause.

- (18) Taamaammat **sisora-neq ajor-pugut** arpa-**tar**-pugut.
therefore ski-NEQ HAB.NEG-1PL.IND run-HAB-1PL.IND
Therefore we don’t ski as much, but run instead.

Ellen Thrane, personal narrative, March 2024.

The example in (19) comes from a social media post. It is clear from the context that the author is conveying that they generally don’t have time for dating, not that there is a particular date that they do not have time for.

⁸The English translations of (17) and (21) are from Bittner’s text collection. Examples (18)–(20) and (22)–(24) were translated into Danish by the Third and Ellen Thrane and then into English by the Line Mikkelsen.

- (19) Aappa-qar-ni-ssa-mut **piffissa-qar-neq ajor-punga** ...
 mate-have-NOM-ALL time-have-NEQ HAB.NEG-1SG.IND
 I don't have time for dating ...

Aleqa Hammond (Facebook)

The PNC is also found children's books as illustrated by the example in (20) taken from *Ataata isertitsivimmiippoq* (Father is in jail.). Here the PNC negates Sofus' expectation that his father would be regularly sending gold back to his family.

- (20) Sofus lives with just his younger brother, older sister and mother. ... Sofus_i misses his_i father_k because he_k has gone to Naluna to dig for gold.

Sofusi-l=li paasi-sinnaa-nngi-laa sooq guldi-nik **nassitsin-neq**
 Sofus-ERG=but understand-can-NEG-3SG>3SG.IND why gold-PL.INST receive-NEQ
ajor-lutik ataata-miniit.
 HAB.NEG-4SG>4PL father-4SG>3SG.ABL

But Sofus can't understand why they [Sofus, his siblings and mother] don't receive gold from his father.

<https://milik.gl/ka/milik/e-bogit/>

The next example comes from a description of the bird species *appaliarsuk* (Little Auk; Alle alle) and its way of life. It is drawn from a passage explaining how it dives for krill. The habitual semantics of the PNC in (21) is conveyed by the use of 'never' in the English translation.

- (21) Qut-sis-su-kkut **ingerla-neq ajor-put.**
 top-be.at-3SG.PART-EQUA travel-NEQ HAB.NEG-3PL.IND
 They [the Little Auks] never fly high
 ... while feeding, first when they've finished feeding do they fly up and go home to roost.
Appaliarsuit (Little auks), §2, (2)

Example (22) comes from the public broadcasting company KNC (Kalaallit Nunaata Radioa; lit. Greenlanders' country's radio) and occurs in a description of a town, whose inhabitants are committed to keeping the town clean. Here the habitual semantics of the PNC is reflected in the English simple present tense translation.

- (22) Maani inuusuttut utoqqa-l=lu maangaannaq igitassa-nik
 here young.person-PL.ABS old-person-PL.ABS=and for.no.reason garbage-PL.INST
igitsi-neq ajor-put, ...
 throw.away-NEQ HAB.NEG-3PL.IND
 Here young and old people don't throw garbage for no reason ...

<https://kni.gl/nunaqarfiga-kusanartog/>

Finally, the PNC is amply attested in newspapers (23) and other non-fiction writing (24).⁹:

⁹In (23) we see the PNC in a generalizing statement. We are not quite sure how to characterize this in terms of habitual semantics, but we include it here to show some of the nuances of the meaning of the PNC.

- (23) Amerlaqi–su–tigut **ilinniaga–qar–neq ajor–put,** ...
 be.many–NOM–VIA.PL education–have–NEQ HAB.NEG–3PL.IND
 Many [children growing up in poverty] don’t get an education, ...
<https://kl.sermitsiaq.ag/inuiagatigiit/inuit-1000-t-sinnerlugit-akerliussutsimik-takutitsipput/957226>
- (24) Sisussat¹⁰ sumiorpaluute–qar–palu–mmat taanna **ator–neq ajor–paat.**
 Sisussat dialect–have–seems–3SG.CAU DEM.DIST use–NEQ HAB.NEG-3PL>3SG.IND
 Because Sisussat seems to have a dialect, they don’t use that [term].
<https://oqaasileriffik.gl/2019/05/20/nunat-aqqinik-aalajangiisartut-2019-1/>

These examples show that the PNC is found in both formal and informal language use and in both traditional language use and current language use. One thing missing from this list of examples above is PNC use in conversation. We currently do not have access to a conversational corpus and also have not had the opportunity to listen for the PNC in conversation. That said, there are reasons to think that the PNC is also used in conversation. First, is it frequent in social media posts, e.g. (19), which is characterized by informality, as is casual conversation. Second, the PNC is sometimes produced in Danish-to-Kalaallisut oral translation tasks in elicitation and also in spontaneous oral narratives, as in (18) above.

3.2 Semantics

As seen in the examples above, the PNC generally expresses habitual negation. This is in contrast with the negative suffix *-nngit*, which expresses ‘plain’ negation and is used in a wide range of semantic contexts, illustrated in the examples below.

First, the suffixal negation example in (25) forms a minimal pair with the PNC example in (17). In (17), the old man is habitually absent (every morning). By contrast, (25) occurs in the telling of a particular episode of the myth *Aataarsuup irnikasia* (Aataarsuaq’s kid son), where Aataarsuaq is in pursuit of his son. Aataarsuaq takes his eyes off his son for just a moment and when he looks again his son is gone.¹¹

- (25) Taamak peqa–**nngi**–laq.
 gone be.there–NEG–3SG.IND.NEG
 Gone, he [Aataarsuaq’s kid son] wasn’t there.
Aataarsuup irnikasia (Aataarsuaq’s kid son), Part IV, §1, (9)

Other representative examples of suffixal negation are included in (26)–(30) below. Collectively these suggest that *-nngit* is a general purpose negator. It can negate various event types, including change of state events (26), semelfactive (29), actions (first clause of (28)) and states (second clause of (28), (30)). It is compatible with modals (27) and can occur in all eight moods, here illustrated with indicative (27), contemporative (26), and conditional (28) moods.

¹⁰This is a local term for a place where a dog sled slides down a steep incline. It also refers to the inhabitants of that place.

¹¹Suffixal negation triggers a special negative allomorph of the following mood-agreement morpheme. This is indicated .NEG in the glossing of the mood-agreement morpheme.

- (26) Suli Tigga-p paata-i ersig-unnaa-**nngi**-llat,
 still alpha.seal-ERG paddle.PL-3SG>3.PL be.visible-cease-NEG-3PL.CONT.NEG
 As yet, Alpha's paddle hasn't become invisible,
Mitinnguaq (Little Eider) §5, (3)
- (27) "Nalus-sinnaa-**nngi**-langa." aapakaaq oqar-poq.
 swim-be.able-NEG-1SG.IND.NEG monkey say-3SG.IND
 "I can't swim.", said the monkey.
Aapakaaq kuukkuuriarlu (The monkey and the crocodile) §4, (5)
- (28) Uku ila-kka ila-a-**nngik**-kaluar-unik suqut-a-anngi-laq.
 these part-1SG>3PL part-be-NEG-FRUS-3PL.COND issue-COP-NEG-3SG.IND.NEG
 If these other pups don't come along, it doesn't matter.
Paakujuk (Sooty) §6, (5)
- (29) Paa-kujuk qilu-ria-qqi-**nngi**-laq.
 soot-be.like bark-stage.one-again-NEG-3SG.IND.NEG
 Sooty didn't give another woof.
Paakujuk (Sooty) §15, (1)
- (30) Toqut-si-so-qar-**ta-nngi**-laq.
 kill-AP-PTCP-have-HAB-NEG-3SG.IND.NEG
 There are usually no murders.

<https://oqaasileriffik.gl/da/sprogteknologi/korpus>

Example (30) is of special interest because it demonstrates that suffixal negation can co-occur with the habitual suffix *tar*, making an alternative strategy for expressing habitual negation available. We have not investigated possible semantic differences between *ta-nngit* (HAB-NEG) and *ajor-*. Our point here is simply that suffixal negation is compatible with habitual semantics, but does not express habituality on its own.

3.3 Status of the PNC

A reviewer asks whether the PNC is marginal in the language. In a qualitative sense, the broad attestation of the PNC documented above suggests that it is not marginal. In terms of frequency, textual counts reveal that the PNC is less significantly frequent than suffixal negation. In the 14 texts curated by Maria Bittner (<https://sites.google.com/view/maria-bittner/kalaallisut>), there are 163 examples of suffixal negation and only 10 examples of the PNC. We don't think this entails that the PNC is marginal in the sense that it belongs at the edge of the grammatical structure of the language. Instead, we would like to suggest, the lower frequency arises from two factors. First, we hypothesize that communicative need to express habitual negation arises less often than the need to express plain negation. If so, it stands to reason that the PNC is less frequent than suffixal negation. The second factor is that there is a morphological alternative to the PNC. As (30) above shows, habitual negation can also be expressed purely morphologically by combining affixal negation with the habitual morpheme *-tar*.

A third way that the PNC could be considered marginal would be if it were a calque from Danish. There are at least two reasons to think the PNC is not a calque. First, Danish has two

ways of expressing habitual negation and neither resemble the PNC. The first strategy is to use the particle used for ‘plain’ negation (*ikke* ‘not’) together with a verb in the simple present tense.

- (31) Hun drikke-**r** **ikke** the om morgen-en.
 she drink-PRES NEG tea about morning-DEF
 She doesn’t drink tea in the morning.

Line Mikkelsen

There is no mapping from this Danish construction to the PNC. In the Danish construction, negation is expressed by a particle, not a verb, and the lexical verb is finite, whereas it is non-finite in the PNC. The second strategy employs the raising verb *pleje* ‘usually do’ in combination with the negative particle.

- (32) Hun **pleje-r** **ikke** at drikke the om morgen-en.
 she usually.do-PRES NEG to drink.INF tea about morning-DEF
 She usually doesn’t drink tea in the morning.

Line Mikkelsen

This construction is a little closer to the PNC in that the lexical verb is non-finite. However, unlike the PNC, negation is expressed as an uninflected particle and the non-lexical verb expresses just habituality, as seen by the fact that it is also used in positive habitual statements (33).

- (33) Hun **pleje-r** at drikke the om morgen-en.
 she usually.do INF drink tea about morning-DEF
 She usually drinks tea in the morning.

Line Mikkelsen.

We thus see no way that either of the Danish constructions could have given rise to the PNC in language contact. Second, the PNC is found in myths, e.g. (17), whose composition presumably predates contact with Danish.

We conclude from the data and discussion in this section that the PNC is relatively infrequent, but not marginal in the grammatical system of the language.

3.4 Homophony of *ajor-*

As noted at the beginning of the paper, the form *ajor-* is also used as an intransitive verb with the meaning ‘be bad’. An example of this use is provided in (34).

- (34) kami-kasi-ni=lu **ajor**-luinna-ler-mata.
 kamik-wretched-4SG>3PL.ABS=and be.bad-absolutely-begin-3PL.CAU
 and because his wretched kamiks had got really bad.

Qillarsuakkunik oqalualaaq (Qillarsuaq’s Saga) §13, (1)

While space prevents us from providing a full discussion, we note that the sentence type in (34) is different from the PNC in at least four ways. First there is no *-neq* verb form involved, just the nominal subject argument ‘his wretched kamiks’. Second, the subject uniformly receives its theta role from *ajor-*. In the PNC, by contrast, the theta role varies with the *neq*-marked lexical verb.¹² Third, lexical *ajor-* is formally strictly intransitive—(34) vs. (35)—whereas the PNC exhibits transitive agreement when the *neq*-marked verb is transitive, e.g. (24).

¹²Here we note in response to a reviewer that the PNC imposes no animacy restrictions on the subject. It is felicitous with non-human (21) and inanimate (i) subjects.

- (35) *Siallu–p Juuna ajor–paa.
rain–ERG Juuna.ABS be.bad–3SG>3SG.IND

Intended: The rain is bad for Juuna.

Grethe Schmidt CZK-GS-2

The last thing to note is the difference in semantics: in the PNC *ajor-* expresses sentence negation and habituality, but not evaluation. In (34) *ajor-* expresses evaluation, but not sentence negation or habituality. The negative auxiliary *ajor-* is clearly synchronically distinct from the lexical intransitive verb *ajor-*. In terms of diachrony we argue in section 7 that auxiliary *ajor-* developed from a transitive verb *ajor-* ‘be unable to reach or do’, which has since fallen out of use. We have not investigated whether intransitive *ajor-* developed from the same transitive negative verb.

4 The duality of *-neq*

In addition to the PNC (36) *neq*-marked verbs feature in control constructions (37).¹³

- (36) Naja **biiler–neq** ajor–poq.
Naja.ABS drive–NEQ HAB.NEG–3SG.IND

‘Naja doesn’t drive.’

Ellen Thrane BPA-3

- (37) Naja–p **biiler–neq** nuannari–vaa.
Naja–ERG drive–NEQ love–3SG>3SG.IND

‘Naja loves to drive.’

Ellen Thrane CHB-4

Despite their surface similarity, we argue that the status of the *neq*-phrase differs. In (36) *biiler–neq* is a non-finite verbal projection. In (37) it is a nominalization.¹⁴ There are three pieces of evidence that the *neq*-phrase is nominal in control structures but not in the PNC.

-
- (i) Ujaqqa–t nakkakat–tar–neq ajor–put.
stone–PL.ABS roll.down–HAB–GER HAB.NEG–3PL.IND

The stones never roll down.

Grethe Schmidt CUL-GS-3

¹³A reviewer asks whether control is necessarily involved in (37) or whether it could ‘simply be a nominalization’. We are not entirely sure what nominalization the reviewer has in mind, but one possibility is a low nominalization of a root, akin to the nominalizing suffix *-al* in English (*refuse* → *refus-al*; Alexiadou (2010)). This analysis, however, is ruled by the fact that *neq*-phrases in both the PNC and control constructions can contain the full suite of argument structure suffixes: passive, antipassive, causative and applicative morphemes (see section 6). If *-neq* were a low nominalizing head that categorizes a root as a nominal, this should not be possible. An alternative interpretation of the reviewer’s suggestion is that it is a larger nominalization structure in which a subject argument is projected but not controlled. The subject inside the *neq*-phrase could instead be *PRO_{arb}* (‘Naja loves that people in general drive’) or existentially closed *pro* (‘Naja loves that someone drives’). These two interpretations predict that (37) can be followed by ‘...but Naja has never driven.’, which it cannot. Instead such statement are reformulated to express now liking to drive, despite not having driven earlier (Grethe Schmidt DAI-GS-2/3). In contrast, the impossibility of the denial follows straightforwardly from a control analysis: the matrix subject *Naja* controls the *PRO* subject of *biiler-* ‘drive’, which entails that what Naja loves is Naja driving. This in turn is incompatible with the second clause that denies her having ever driven. Finally note that our analysis of the PNC is unaffected by which type of nominalization is involved in (37); all that matters for the comparison of the two constructions is that (37) involves some kind of nominalization. For recent discussion of control into nominalizations see Mourounas (2019).

¹⁴This duality of V-*neq* is reminiscent of the duality of English *-ing* forms, which also have verbal and nominal guises (Abney 1987:167–244). The PNC most closely resembles the participle use of *-ing* as in *Naja is singing silly*

First, in control structures, the *neq*-phrase can bear possessive morphology (38); in the PNC it cannot, whether or not *ajor*- bears intransitive (39) or transitive (40) agreement.

- (38) Naja-p mersor-ni-**ni** nuannari-vaa.
 Naja-ERG sew-NEQ-4SG>3SG.ABS love-3SG>3SG.IND
 ‘Naja_i loves her_i sewing.’ Ellen Thrane CHR-1
- (39) *Naja mersor-ni-**ni** ajor-poq.
 Naja.ABS sew-NEQ-4SG>3SG.ABS HAB.NEG-3SG.IND
 Intended: ‘Naja doesn’t sew.’ Ellen Thrane CHR-2
- (40) *Naja-p mersor-ni-**ni** ajor-paa.
 Naja-ERG sew-NEQ-4SG>3SG.ABS HAB.NEG-3SG>3SG.IND
 Intended: Naja doesn’t sew. Ellen Thrane CII-7

Second, control verbs like *nuannari*- ‘love’ can take underived nominal complements (41), the negative auxiliary *ajor*- cannot (42):

- (41) Sialuk nuannari-vara.
 rain.ABS love-1SG>3SG.IND
 ‘I love rain.’ Grethe Schmidt CKJ-GS-2
- (42) *Sialuk ajor-poq.
 rain.ABS HAB.NEG-3SG.IND
 Intended: ‘It doesn’t rain.’ Grethe Schmidt CKJ-GS-1

These differences in possessive morphology and subcategorization fall into place if *V-neq* is verbal in the PNC but nominal in control structures.

A third potential contrast between control constructions and the PNC comes from the claim in Sadock (2003:10, 33) that in the PNC, the *neq*-marked verb must be immediately adjacent to the negative auxiliary. This is clearly not the case for control verbs. Examples like (43), where the *neq*-phrase has scrambled over the subject, are always judged grammatical.

- (43) Mersor-neq Naja-p nuannari-vaa.
 sew-NEQ Naja-ERG love-3SG>3SG.IND
 ‘Naja loves to sew.’ Grethe Schmidt CIC-GS-2

This word order difference would follow from the *neq*-phrase being a DP in control structures but a TP in the PNC, since DPs can scramble but TP cannot (Grewendorf and Sternefeld 1990:13–14; Webelhuth 1992:180). However, in our work, scrambling of the *neq*-phrase in a PNC receives mixed judgments.

- (44) %Mersor-neq Naja ajor-poq.
 sew-NEQ Naja.ABS HAB.NEG-3SG.IND
 Intended: ‘Naja doesn’t sew.’

songs. The Kalaallisut control construction in (37), however, has no correspondent in English. In Abney’s terms it is a blend of *PRO-ing* (*Naja_i likes PRO_i singing silly songs*) and *poss-of-ing* (*Naja likes Harvey’s singing of silly songs*). That blend is ungrammatical in English: **Naja_i likes PRO_i singing of songs*, suggesting that the analogy between Kalaallisut *-neq* and English *-ing* is not exact. While we agree that this contrast between nominal *-ing/-neq* constructions is worthy of further exploration, we will not attempt it here.

The Grethe Schmidt rejected examples like (44) on four out of six occasions and the Ellen Thrane rejected it on three out of five occasions. One possible interpretation of this state of affairs is that the ban on scrambling of TP, which was established mostly based on languages with a very different typological profile, is weaker in Kalaallisut and manifests as TP scrambling being of variable acceptability. Under that interpretation, the relative acceptability contrast between (43) and (44) supports our claim that in control structures the *neq*-phrase is a DP, whereas it is verbal in the PNC.

Anticipating our analysis in section 6, we propose that in the PNC *-neq* heads a non-finite verbal projection, which we identify as TP. This straightforwardly explains its verbal behavior in the PNC: it does not host possessive morphology, it does not alternative with DP complements, and scrambling is of variable acceptability. In control structures, *-neq* has its usual nominalizing function, which we take to be the manifestation of a D head.¹⁵ Being a DP, the *neq*-phrase in a control construction can host nominal morphology, alternates with a non-derived DP complement, and scrambles.

Before we turn to a more detailed examination of case and agreement properties of PNC and control structures, we briefly consider past treatments of the *neq*-suffix. All descriptions that we are aware of treat *-neq* as a nominalizing suffix, including Bittner (1994:53ff), Fortescue (1984:44ff), Nielsen (2021:199) and Olsen and Hertling (2011:48). More specifically Bittner (1994:53-54, 64) analyzes *-neq* as a noun in her account of control into adjunct clauses formed with *-neq* (see (55)). While this analysis can be extended to account for the nominal properties of *neq*-phrases in the complement control structures we are concerned with here, it doesn't extend to *neq*-phrases in the PNC, which have none of these nominal characteristics. In sum, while some past descriptions of *-neq* have noted the PNC and its resemblance to complement control structures, none of them have provided an analysis that recognizes their differences when it comes to the function of *-neq*.

5 Case and agreement

The discussion above lets us posit the following schematic structures for the PNC (45) and control (46):¹⁶

- | | | |
|------|---|---------|
| (45) | Subj [_{TP} VP- <i>neq</i>] Aux | PNC |
| (46) | Subj [_{DP} VP- <i>neq</i>] V | CONTROL |

Now we examine the predictions about case and agreement that follow from these structures together with general principles of case and agreement in the language. We start with control.

¹⁵Another analytic possibility is that *-neq* is a n head in control constructions (see Alexiadou (2020) for an overview of nominalization by D and by n). We will not attempt to argue n vs. D nominalization here, as the only thing that matters for us is that there is nominalization in control structures but not in the PNC. See Carrier and Compton (2024) for relevant discussion of the category of the cognate Inuktitut nominalizer *-niq*.

¹⁶In (45) and (46) VP stands in for the argument structure projections vP, Pass(ive)P, Caus(ative)P, Appl(icative)P, and a(anti)p(assive)P; see section 6.

5.1 Control

Our starting point for control structures is that the *neq*-phrase complement of the control verb is a DP. This makes five predictions about case and agreement which we consider in turn.

Prediction 1 As an internal DP argument of a transitive verb, the *neq*-phrase is absolutive. This claim is hard to assess in basic control constructions because absolutive case is null. However, possessive *neq*-DPs show distinctive absolutive morphology shared with possessed underived nominals (47) and (48). This absolutive possessive morphology contrasts with possessive morphology in other cases, including ergative possessive morphology in (49).¹⁷

- (47) Hansi-p qimmi-**ni** nuannari-vaa.
Hansi-ERG dog-4SG>3SG.ABS love-3SG>3SG.IND
‘Hansi_i loves his_i dog.’ Grethe Schmidt CIH-GS-1
- (48) Naja-p mersor-ni-**ni** nuannari-vaa.
Naja-ERG sew-NEQ-4SG>3SG.ABS love-3SG>3SG.IND
‘Naja loves to sew./Naja_i loves her_i sewing’ Ellen Thrane CHR-1
- (49) Hansi-p qimmi-**ata** kii-vaanga.
Hansi-ERG dog-4SG>3SG.ERG bite-3SG>1SG.IND
‘Hansi’s dog bit me.’ Grethe Schmidt CIH-GS-2

Prediction 2 As the subject of a verb with an absolutive object, the subject is ergative (Yuan 2018). This holds true even if the embedded verb is intransitive, as seen in (50).

- (50) **Naja-p/*Naja** biiler-neq nuannari-vaa.
Naja-ERG/Naha.ABS drive-NEQ love-3SG>3SG.IND
‘Naja loves to drive.’ Ellen Thrane CHB-4/CHB-5

Prediction 3 Since there is at most one absolutive DP per CP (Compton 2018:164), it should not be possible to realize an internal argument of the *neq*-marked verb as an absolutive. This is indeed impossible:

- (51) *Meeqqa-nut poortukka-t nassi-un-neq nuannari-vara.
child-PL.ALL package-PL.ABS send-APPL-NEQ love-1SG>3SG.IND
Intended: I love to send packages to the children. Ellen Thrane CKI-3
- (52) *Iipili-t neri-neq qatsup-para.
apple-PL.ABS eat-NEQ be.tired.of-1SG>3SG.IND
Intended: ‘I am tired of eating apples.’ Grethe Schmidt CIE-GS-4

¹⁷The possessive inflections on examples (47)-(49) are anaphoric in the sense that they obligatorily reference the subject. As noted above, there is a second series of inflections for disjoint reference between the possessor and a higher subject. Disjoint possessive inflection on the *neq*-phrase is impossible in both the PNC and control constructions. For the PNC the impossibility of disjoint possessive morphology follows from the *neq*-phrase being verbal. For control structures it follows from the possessive morphology referencing the subject.

Instead such internal arguments are either incorporated (53) or realized as an oblique, e.g. an instrumental object of an antipassivized verb (54a) or an instrumental object of a notionally transitive, but formally intransitive verb form (54b); see also Fortescue (1984:45).

- (53) a. Puisi–tor–neq nuannara–arput.
 seal–consume–NEQ love–1 PL>3 SG.IND
 ‘We love to eat seal meat.’ Ellen Thrane BSY-2
- b. Iipili–tor–neq qatsup–para.
 apple–consume–NEQ be.tired.of–1 SG>3 SG.IND
 ‘I am tired of eating apples.’ Grethe Schmidt CIE-GS-6
- (54) a. Meeqqa–nik iku–i–neq nuannari–vara.
 child–PL.INST help–AP–NEQ love–1 SG>3 SG.IND
 ‘I love to help the children.’ Grethe Schmidt CHC-GS-2
- b. Iipili–nik neri–neq qatsup–para.
 apple–PL.INST eat–NEQ be.tired.of–1 SG>3 SG.IND
 ‘I am tired of eating apples.’ Grethe Schmidt CIE-GS-5

Prediction 4 If a control construction makes an oblique case available, the *neq*-phrase should be able to expone that case. We see this in the example in (55), where the matrix verb ‘be tired’ licenses a source argument. That source argument is realized with ablative case on the *neq*-phrase. Because the *neq*-phrase is not absolutive, the internal argument of *malisor*- ‘chase’ can be realized as an independent absolutive DP without running afoul of the single absolutive restriction. Compare (55) to (51) and (52).¹⁸

- (55) Juuna–p qilalukka–t malisor–ner–mit qasu–voq.
 Juuna–ERG whale–PL.ABS chase–NEQ–ABL be.tired–3 SG.IND
 ‘Juuna is tired from chasing whales.’ Bittner (1994:53), ex. 9a

Prediction 5 As an absolutive argument of a transitive verb, the *neq*-DP should control object agreement on the matrix control verb. As a nominalization, the ϕ -features of the *neq*-DP are 3SG (e.g. Hanink 2021:545-6, Picallo 2001). We thus correctly predict obligatory 3SG object agreement in control constructions.

¹⁸Carrier and Compton (2024) investigate a similar contrast in Inuktitut nominalizations involving a cognate suffix *-niq*. These *-niq* structures allow an internal absolutive argument when occurring in adjunct position, but not when occurring in argument position. Carrier and Compton (2024) attribute this contrast to a size difference between adjunct and argument *niq*-nominalizations. Adjunct nominalizations are large enough to license absolutive case, specifically by Force and Topic heads above *-niq*. Argument nominalizations lack the Force and Topic projections and hence absolutive case is not licensed. While we have not investigated the distribution of Force and Topic heads in Kalaallisut, it seems quite possible to us that Carrier and Compton’s analysis could extend to the Kalaallisut control constructions that we are comparing the PNC to. The *neq*-phrase is in an argument position (it is a complement to the control verb) and as such should lack Force and Topic projections without which no absolutive case is available for an internal argument of the embedded verb. The key point for us is that *neq*/*niq* are nominalizers in argument positions. That is what sets them apart for *neq*-phrases in the PNC.

- (56) Mersor–neq nuannari–**vara**.
 sew–NEQ love–1SG>3SG.IND
 ‘I love to sew.’

Ellen Thrane BTM-3

- (57) *Mersor–neq nuannari–**vunga**.
 sew–NEQ love–1SG.IND
 Intended: ‘I love to sew.’

Ellen Thrane BTM-2

An alternative interpretation of (56) is that the 3SG object agreement is a form of default agreement. There are two reasons to resist that interpretation. First, coordinated *neq*-phrases can trigger 3PL object agreement:

- (58) Erinarsor–neq qitin–ner=lu nuannari–vakka
 sing–NEQ dance–NEQ–CONJ love–1SG>3PL.IND
 ‘I love to sing and dance.’

Ellen Thrane DJE-5

This is direct evidence for a genuine number feature on the *neq*-phrase and unexplained on a default agreement analysis. Second, we don’t see default object agreement in other contexts where person-number features of a notional object are inaccessible for agreement. Recall that verbs only agree with ergative and absolutive arguments. As noted in the discussion of (54b) above, there is a class of notionally transitive verbs, including *neri*- ‘eat’, which occur in two case frames: ERG-ABS and ABS-INST.^{19,20}

In the ERG-ABS case frame, the verb shows transitive agreement:

- (59) Naja–p iipili–t pingasut neri–vai.
 Naja–ERG apple–PL.ABS three.ABS eat–3SG>3PL.IND
 ‘Naja ate three apples.’

Grethe Schmidt CKL-GS-1

Since verbs can’t agree with oblique arguments, we might expect default 3SG object agreement in the ABS-INST frame. But that is impossible (60). Instead we find intransitive agreement (61).

- (60) *Naja iipili–nik pingasu–nik neri–vaa.
 Naja.ABS apple–INST.PL three–INST.PL eat–3SG>3SG.IND
 Intended: Naja ate three apples.

Grethe Schmidt CKL-GS-3

- (61) Naja iipili–nik pingasu–nik neri–voq.
 Naja.ABS apple–INST.PL three–INST.PL eat–3SG.IND
 Naja ate three apples.

Grethe Schmidt CKL-GS-4

¹⁹As pointed out by a reviewer, the ABS-INST construction in (54b) and (61) could also be analyzed as involving a null antipassive morpheme on the verb. Our argument against default case goes through under either analysis.

²⁰According to traditional treatments like those in Kleinschmidt (1851/1968), Bergsland (1955), Woodbury (1975), Fortescue (1984), and Sadock (1980), absolutive objects are definite and instrumental objects are indefinite. Bittner (1987) shows that the situation is considerably more complex and argues that the relevant difference is one of scope: absolutive objects take wide scope, whereas instrumental objects must take narrow scope. In our work there is not a consistent correlation between an absolutive case-marked object and a definite translation nor between an instrumental case-marked object and an indefinite translation. Since we are not concerned with semantic interpretation of the constructions under discussion, we set this issue aside and simply provide faithful English translations of the Danish translations established in elicitation.

In other words, outside of control structures, there is no evidence for default agreement. In fact there is evidence against it from the ungrammaticality of (60). We conclude that control verbs do agree with their *neq*-phrase complement, lending support to our claim that the *neq*-phrase is nominal in these structures.

5.2 Periphrastic negation structures

For the PNC, our hypothesis is that V-*neq* is a non-finite verbal projection, specifically a non-finite TP. From this hypothesis four predictions follow.

Prediction 1 As a TP, the *neq*-phrase cannot bear any case morphology, be it a possessive absolutive case (62), instrumental case (63) or allative case (64). (While not shown for reasons of space the other five cases—ablative, equative, ergative, locative, and prolative—are equally impossible on the *neq*-phrase in the PNC.)

- (62) *Naja-p mersor-ni-ni ajor-paa.
 Naja-ERG sew-AP-4SG>3SG.ABS HAB.NEG-3SG>3SG.IND
 Intended: Naja loves to sew. Ellen Thrane CII-7
- (63) *Naja biiler-nim-mik ajor-poq.
 Naja.ABS drive-NEQ-INST HAB.NEG-3SG.IND
 Intended: Naja doesn't drive. Grethe Schmidt BRF-GS-4
- (64) *Meeqqa-t ikior-nim-mut ajor-pakka.
 child-PL.ABS help-NEQ-ALL HAB.NEG-1SG>3PL.IND
 Intended: I didn't help the children. Grethe Schmidt BRG-GS-3

Absolutive and instrumental case are the most obvious cases to consider. Absolutive is relevant because the *neq*-phrase is the complement of a finite verbal form, as is the case for absolutive DPs. Instrumental is relevant because it alternates with instrumental case in antipassive constructions yielding intransitive agreement on the verb, which is associated with an intransitive *neq*-complement to *ajor-*, here 'drive'. The allative case example in (64) is ungrammatical, as are all other oblique cases.

Prediction 2 Without an absolutive internal argument of the lexical verb, the subject cannot be ergative, whether the negative auxiliary shows transitive or intransitive agreement:

- (65) *Naja-p biiler-neq ajor-paa/poq.
 Naja-ERG drive-NEQ HAB.NEG-3SG>3SG.IND/3SG.IND
 Intended: Naja doesn't drive. Grethe Schmidt CIH-GS-4/CIH-GS-3

Prediction 3 Since the *neq*-phrase is not absolutive, absolutive case is available for internal arguments of the lexical verb, as seen for *meeqqa-ni* 'her children' in (66).

- (66) Naja-p meeqqa-ni ikior-neq ajor-pai.
 Naja-ERG child-4SG>3SG.ABS help-NEQ HAB.NEG-3SG>3PL.IND
 'Naja_i didn't help her_i children.' Grethe Schmidt BRJ-GS-2

Prediction 4 As the *neq*-phrase is not absolutive, it cannot control object agreement on the negative verb:

- (67) *Biiler–neq ajor–para.
 drive–NEQ HAB.NEG–1SG>3SG.IND
 Intended: ‘I don’t drive.’

Grethe Schmidt CHA-GS-2

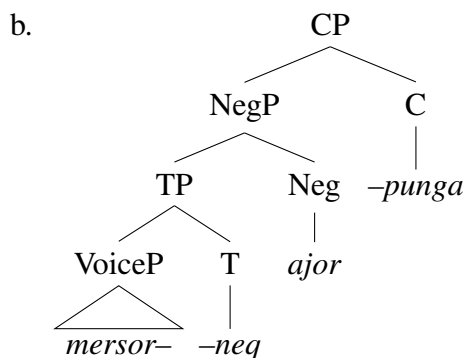
To summarize: in the PNC the *neq*-phrase is syntactically inert as regards case and agreement. It does not bear case itself, it does not license ergative case on the higher subject, it is transparent for absolutive case assignment, and it does not control agreement. This follows straightforwardly on the hypothesis that in the PNC the *neq*-phrase is exclusively verbal.

6 The structure of *neq*- constructions

In this section we provide phrase-structural analyses of the PNC and control structures, based on the observations made above. As the analysis revolves around the syntactic status of the *neq*-suffix, we build our analysis from an examination of the position of *-neq* relative to the position of argument structure morphology and inflectional morphology. We show that in both the PNC and control structures the *neq*-head is positioned between argument structural projections (labeled VoiceP in (68b) and (69b)) and inflectional heads (section 6.1). For control verbs, nothing more needs to be said. For the PNC, we make the further argument that *ajor*- is a Neg head positioned above T (section 6.2). The two analyses are illustrated in the trees below; following Compton (2018), we take mood and agreement to be exponed in C.

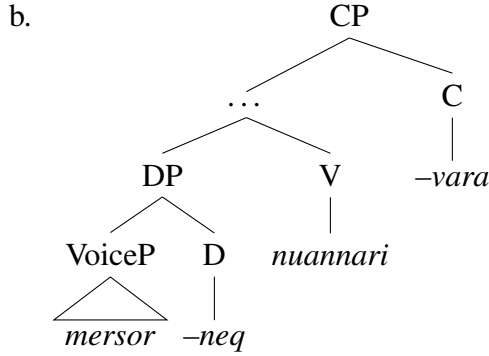
- (68) a. Mersor–neq ajor–punga.
 sew–NEQ HAB.NEG–1SG.IND
 ‘I don’t sew.’

Ellen Thrane CJP-4



- (69) a. Mersor–neq nuannari–vara.
 sew–NEQ love–1SG>3SG.IND
 ‘I love to sew.’

Grethe Schmidt BTR-GS-1



6.1 Structure below *-neq*

Before we document the syntactic material that *-neq* can embed we need to say a few words about the clausal structure we assume for Kalaallisut and about the scope of our investigation. Morphology reveals that the following heads can appear on the verbal spine: i) V which corresponds to the verbal root and introduces any internal arguments; ii) Caus which introduces a causer argument and demotes the embedded Agent to oblique, iii) Appl which promotes an oblique argument to direct object; iv) the antipassive head *ap* (Yuan 2018:205ff), which demotes the notional object to an oblique, and v) Voice, which introduces the Agent in active clauses and hosts the passive suffix in passive clauses. We assume that these heads are only present when warranted by syntactic material in a particular construction. That is, we are not adopting a cartographic approach. We will not be concerned with the relative height of these heads, but simply document that any and all of them can occur inside the *neq*-phrase. (For ordering of the corresponding derivational morphemes see Fortescue (1980).) We also limit ourselves to argument-structure changing morphology and indisputably inflectional morphology, which is mood and agreement. We do this for two reasons. First, there are 300–400 derivational suffixes that occur in Kalaallisut verbs (Fortescue 1980:259) and it would be practically impossible to examine for each of them whether it can occur inside the *neq*-phrase. Second, the suffixes that we choose to look at are exactly the ones that are most informative for our analytic purpose. Argument-changing morphology interacts with case and agreement, which is what we are analyzing. Inflectional morphology is hosted in C and the very outermost morphology of the verb word. As such, it let's us determine whether the upper bound of *-neq* is below the finite CP.

Our documentation relies on the Mirror Principle which states that “[m]orphological derivations directly reflect syntactic derivations (and vice versa)” (Baker 1985:375). Of particular relevance is its corollary regarding morpheme order and the hierarchy of syntactic projections: if morpheme α occurs closer to the root than morpheme β , then the syntactic head responsible for α is lower in the syntactic structure than the syntactic head responsible for β . We will see that any argument-structure changing morphology can occur closer to the root than *-neq*, which indicates that VP, *ap*P, CausP, ApplP and VoiceP can all be projected below the *neq*-head. In the PNC this is true in all circumstances, whereas it is modulated by the ban on multiple absolutes in control structures. In neither construction is inflectional morphology inside *-neq* possible.

We start our examination with basic active clauses. As seen above already, these are licit in both the PNC and in control structures.

(70) ACTIVE VOICE

- a. Naja-p meeqqa-t ikior-neq ajor-pai
 Naja-ERG child-PL.ABS help-NEQ HAB.NEG-3SG>3PL.IND
 ‘Naja didn’t help the children.’ Grethe Schmidt BRI-GS-2
- b. Naja-p mersor-neq nuannari-vaa.
 Naja-ERG sew-NEQ love-3SG>3SG.IND
 ‘Naja loves to sew.’ Grethe Schmidt CIB-GS-2

neq-phrases formed on passives are also possible in both constructions, with the passive suffix *-neqar* occurring inside *-neq* :

(71) PASSIVE VOICE

- a. Meeqqa-t Naja-mik ikior-**neqar**-neq ajor-put.
 child-PL.ABS Naja-INST help-PASS-NEQ HAB.NEG-3PL.IND
 ‘The children are not being helped by Naja.’ Ellen Thrane BRN-7
- b. Meeqqa-t Naja-mik ikior-**neqar**-neq nuannari-vaat.
 child-PL.ABS Naja-INST help-PASS-NEQ love-3PL>3SG/PL.IND
 ‘The children love to be helped by Naja.’ Ellen Thrane CIP-1

The antipassive morpheme *-i* likewise occurs inside *-neq* indicating that the antipassive phrase, apP, can also be contained in the *neq*-phrase:

(72) ANTIPASSIVE

- a. Naja meeqqa-nik ikiu-**i**-neq ajor-poq.
 Naja.ABS child-INST.PL help-AP-NEQ HAB.NEG-3SG.IND
 ‘Naja doesn’t help the children.’ Ellen Thrane BRP-1
- b. Naja-p meeqqa-nik ikiu-**i**-neq nuannari-vaa.
 Naja-ERG child-PL.INST help-AP-NEQ love-3PL>3SG.IND
 ‘Naja loves to help the children.’ Ellen Thrane CIP-3

The causative suffix *-tit* can occur inside *-neq* in the PNC (73).²¹

- (73) Juuna-p Naja-mut meeqqa-t ikior-**tin**-neq ajor-pai.
 Juuna-ERG Naja-ALL child-PL.ABS help-CAUS-NEQ HAB.NEG-3SG>3PL.IND
 ‘Juuna doesn’t let Naja help the children.’ Ellen Thrane CIK-1

In contrast, causative *-tit* inside *-neq* is ungrammatical in the control example in (74).

- (74) *Juuna-p Naja-mut meeqqa-t ikior-**tin**-neq nuannari-vaa.
 Juuna-ERG Naja-ALL child-PL.ABS help-CAUS-NEQ love-3SG>3SG.IND
 Intended: ‘Juuna loves letting Naja help the children.’ Ellen Thrane CFK-11

²¹As discussed by Fortescue (1984:84ff), *-tit* allows a reading akin to an analytic ‘let’-causative, as in (73). Also note that final consonant of the causative morpheme assimilates to the initial consonant of *-neq* producing the surface form *-tin-neq*. Here we use *-tit* as the citation form following Fortescue (1984:84)

We attribute this contrast to the single absolutive restriction: in (73) there is only one absolutive DP, namely the *meeqqa-t* ‘children’. In (74) there are two: *meeqqa-t* and the nominalized DP that contains it ([_{DP} *Naja-mut meeqqa-t ikior-tin-neq*] ‘letting Naja help the children’). Evidence that it is indeed the single absolutive restriction that is at play and not a difference in whether the *neq*-phrase can embed CausP in the two constructions comes from the example in (75). This is an antipassivized version of (74). As such the theme is not absolutive, but instrumental and the entire construction contains only one absolutive, namely the DP complement of the control verb. The grammaticality of (75) thus confirms that the *neq*-head can embed CausP in a control construction, just as it can in the PNC.

- (75) Juuna-p Naja-mut meeqqa-nik iku-i-**tin**-neq nuannari-vaa.
 Juuna-ERG Naja-ALL child-PL.INST help-AP-CAUS-NEQ love-3SG>3SG.IND
 ‘Juuna loves letting Naja help the children.’ Ellen Thrane CKP-1

The applicative morpheme *-ut* has multiple functions (Woodbury 1977), two of which are relevant here.²² First, it transitivizes an intransitive verb (76). Second, when applied to a ditransitive verb, it promotes an oblique theme to direct object, as seen in (77b). (The underived form in (77a) shows that Kalallisuut has secundative alignment in basic ditransitive structures.)

- (76) TRANSITIVIZING FUNCTION OF -UT
- a. Ummi-vunga.
 aim-1SG.IND
 ‘I took aim.’ Ellen Thrane CKQ-1
- b. Orpi-it ummi-**up**-pakka.
 tree-PL.ABS aim-APPL-1SG>3SG.IND
 ‘I aimed for the trees.’ Ellen Thrane CKQ-2
- (77) THEME PROMOTING FUNCTION OF -UT
- a. Meeqqa-t poortukka-nik nassip-pakka.
 child-PL.ABS package-PL.INST send-1SG>3PL.IND
 I sent the children the packages. Ellen Thrane CIM-1
- b. Meeqqa-nut poortukka-t nassi-**up**-pakka.
 child-PL.ALL package-PL.ABS send-APPL-1SG>3PL.IND
 I sent the packages to the children. Ellen Thrane CIM-2

The PNC allows both applicative structures to surface inside the *neq*-phrase:

- (78) a. Orpi-it ummi-**un**-neq ajor-pakka.
 tree-PL.ABS aim-APPL-NEQ HAB.NEG-3SG>3PL.IND
 ‘I don’t aim for the trees.’ Ellen Thrane CKQ-4
- b. Meeqqa-nut poortukka-t nassi-**un**-neq ajor-pakka.
 child-PL.ALL package-PL.ABS send-APPL-NEQ HAB.NEG-1SG>3PL.IND
 ‘I didn’t send the package to the children.’ Grethe Schmidt CJM-GS-1

²²Like causative *-tit*, the final consonant of *-ut* assimilates to a following consonant.

In the control structure, we again see the effect of the single absolutive restriction. Specifically, the structures in (79) are ungrammatical because in addition to the absolutive *neq*-DP, there is a second absolutive DP (*orpiit* in (79a) and *poortukkat* in (79b)).

- (79) a. *Orpi-it ummi-**un-neq** qatsup-para.
 tree-PL.ABS aim-APPL-NEQ be.tired.of-1SG>3SG.IND
 Intended: ‘I am tired of aiming for the trees.’ Ellen Thrane CKR-2
- b. *Meeqqa-nut poortukka-t nassi-**un-neq** nuannari-vara.
 child-PL.ALL package-PL.ABS send-APPL-NEQ love-1SG>3SG.IND.
 Intended: ‘I love to send packages to the children.’ Ellen Thrane CKI-3

Again we can show that the issue is not embedding ApplP under the *neq*-head, but instead the ban against multiple absolutives. Starting with the ungrammatical applicative structure in (79a), we can avoid multiple absolutives by antipassivizing the verb so that the theme is realized with instrumental case. This strategy is confirmed by the grammaticality of (80a). Similarly, we can antipassivize the ditransitive verb in (79b) to produce the grammatical (80b); the final /p/ of the causative suffix assimilates to the initial /s/ of the antipassive morpheme.

- (80) a. Orpin-nik ummi-**us-si-neq** qatsup-para.
 tree-PL.INST aim-APPL-AP-NEQ be.tired.of-1SG>3SG.IND
 ‘I am tired of aiming for the trees.’ Ellen Thrane CKR-8
- b. Meeqqa-nut poortukka-nik nassi-**us-si-neq** nuannari-vara.
 child-PL.ALL package-PL.INST send-APPL-AP-NEQ love-1SG>3SG.IND.
 ‘I love to send packages to the children.’ Ellen Thrane CLO-3

In summary, the *neq*-phrase can embed active and passive VoiceP, ApplP, CausP and apP. In contrast *neq*-phrases verbs cannot contain inflectional morphology, specifically mood and agreement morphology, in neither the PNC nor control structures. The most likely mood to be possible is contemporative mood, which is used when the subject of an embedded clause is coreferential with the matrix subject, as in (81).

- (81) Naja oqar-poq [_{CP} tamatigut angerla-jaar-**luni**].
 Naja.ABS say-3SG always leave-early-4SG.CONT
 Naja_i said that she_i always leaves early. Ellen Thrane BCC-2

In control constructions the embedded subject is likewise coreferential with the matrix subject (Naja_i loves PRO_i to drive) and thus if any mood is available it would be the contemporative mood. Yet, contemporative mood inside the *neq*-phrase in a control construction is impossible as (82) shows.

- (82) *Naja-p biiler-**luni-neq** nuannari-vaa.
 Naja-ERG drive-4SG.CONT-NEQ.ABS love-1SG>3SG.IND
 Intended: ‘Naja loves to drive.’ Ellen Thrane CIR-4

Contemporative mood morphology inside the *neq*-phrase in the PNC is similarly ungrammatical:

- (83) *Naja biiler-**luni**-neq ajor-poq.
 Naja.ABS drive-4SG.CONT-NEQ HAB.NEG-3SG.IND
 Intended: ‘Naja doesn’t drive.’ Ellen Thrane CIR-2

Unsurprisingly, the other seven moods—indicative, interrogative, optative, participial, causative, conditional, imperative—are also impossible inside *-neq* in both control structures and the PNC. Here we illustrate with indicative (84) and participial moods (85).

- (84) a. *Naja biiler-**por**-neq ajor-poq.
 Naja.abs drive-3SG.IND-NEQ HAB.NEG-3SG.IND
 Intended: Naja doesn’t drive. Ellen Thrane CIR-1
- b. *Naja-p biiler-**por**-neq nuannari-vaa.
 Naja-ERG drive-3SG.IND-NEQ love-1SG>3SG.IND
 Intended: ‘Naja loves to drive.’ Grethe Schmidt DAM-GS-2
- (85) a. *Naja biiler-**tor**-neq ajor-poq.
 Naja.ABS drive-3SG.PART-NEQ HAB.NEG-3SG.IND
 Intended: Naja doesn’t drive. Grethe Schmidt DAL-GS-1
- b. *Naja-p biiler-**tor**-neq nuannari-vaa.
 Naja-ERG drive-3SG.PART-NEQ.ABS love-1SG>3SG.IND
 Intended: ‘Naja loves to drive.’ Grethe Schmidt DAL-GS-3

The data provided in this section are summarized in the table below. Recall that for the causative and applicatives, the control construction requires antipassivization to not run afoul of the single absolutive restriction.

(86) POSSIBLE STRUCTURE BELOW -NEQ

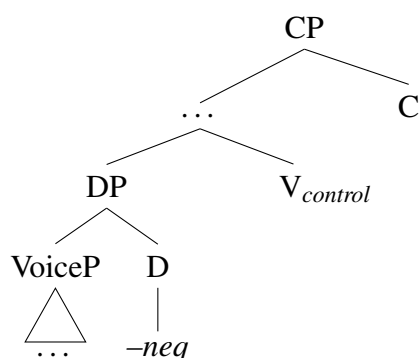
STRUCTURE	PNC	CONTROL
active	✓	✓
passive	✓	✓
antipassive	✓	✓
applicative of intransitive	✓	✓
applicative of transitive	✓	✓
causative	✓	✓
agreement/mood	✗	✗

On our structural analyses of the PNC and the control construction, this distribution is due to the *neq*-head invariably taking VoiceP as its complement and never CP, as would be required for mood and agreement morphology to occur inside *-neq*. In section 7, we return to the observation that the *neq*-head embeds the same amount of structure in the two constructions and suggest that it stems from the PNC having developed from a control structure.

6.2 Structure above the *neq*-phrase

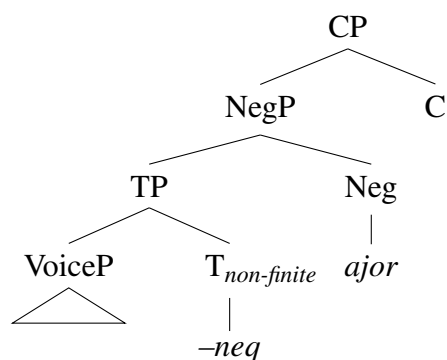
In control structures *-neq* heads a DP, which in turn occupies the complement position of the control verb.

(87) CONTROL



In the PNC, the *neq*-phrase is a non-finite TP complement of the negative head *ajor*-. Following Holmberg et al.'s (1993:183–185) analysis of negative auxiliaries in Finnish (ISO 639: fin), we propose that *ajor*- projects a NegP as in (88).²³

(88) PNC



Evidence for a NegP analysis comes from the incompatibility of the negative auxiliary and suffixal negation *-nngit*:²⁴

- (89) *Naja biiler-neq **aju-nngi**-laq.
 Naja.ABS drive.car-NEQ HAB.NEG-neg-3SG.IND.NEG
 Intended: 'Naja drives. [lit. Naja doesn't not drive]'

Grethe Schmidt BSR-GS-4

We note that the lexical verb *ajor*- 'be bad' can co-occur with suffixal negation as in the example in (90).

- (90) pisiari-gukku **aju-ssa-nngi**-laq
 buy-2SG>3SG.COND be.bad-FUT-NEG-3SG.IND.NEG
 'It would be good if you bought it.'

Fortescue 1984:33

²³This analysis does not, in and of itself, explain the habitual meaning of the PNC. A reviewer suggests that *-ajor* could be a single morpheme realization of a span of habitual Asp and Neg (Svenonius, 2012). We find this idea appealing, but have chosen to leave it for future research, partly because we have not investigated whether Asp and Neg form a span in the PNC, partly for reasons of space.

²⁴There are three morpho-phonological processes at play in (89). First the negative suffix deletes the final /t/ of the negative auxiliary. This in turn bleeds the phonological rule that lowers /u/ to /o/ before /t/. Third *-nngit* triggers the special allomorph of the 3SG.IND morpheme, *-laq* in place of the regular form *-poq*.

In accord with the ungrammaticality judgment in (89), we have not found any examples of negative auxiliary *ajor* and *-nngit* cooccurring in Bittner’s text collection or in Fortescue (1984), Sadock (2003), and Kahn and Valijärvi (2022). In addition to being in complementary distribution with *-nngit*, both morphemes are followed by mood and agreement morphology as seen in (91).

- (91) a. Biili–**nngi**–**langa**.
drive–NEG–1 SG.IND.NEG
‘I don’t drive.’ Grethe Schmidt CMB-GS-2
- b. biiler-neq **ajor**–**punga**.
drive–NEQ HAB.NEG–1 SG.IND
‘I don’t drive.’ Grethe Schmidt CMB-GS-3

The proposed structure also explains the ungrammaticality of (92):

- (92) *Naja biiler-neq **aju**–**ssa**–aq.
Naja drive.car-NEQ HAB.NEG-FUT-3SG.IND
Intended: Naja will not drive in the future. Grethe Schmidt BSF-GS-1

In this example future tense is stacked on top of *ajor*- which doesn’t fit the structure in (88), where T is below Neg.

Moreover, in its habitual negation use, *ajor*- does not stand in a thematic relationship to any of the DPs it co-occurs with—i.e. *ajor*- does not assign any theta roles—suggesting that it is not a lexical verb. Finally, the fact that *ajor*- contributes negation makes it a good fit for Neg⁰. Our conclusion is therefore that *ajor*- projects a NegP, as in (88).

7 Diachrony

Above we have argued that the PNC differs from control constructions in two key respects: the *-neq* phrase is verbal in the PNC, but nominal in control constructions. Second, *ajor*- is an auxiliary, whereas control verbs like *nuannari*- (‘love to’) is a lexical verb. In this section we sketch a diachronic analysis of how these differences came to be. In short we propose that the auxiliary *ajor*- developed from a lexical control verb *ajor*- in tandem with a change in the category of the *-neq* phrase from nominal to verbal. Both are well-established pathways of change. Heine and Kuteva (2002) and Heine et al. (2019) list numerous examples of lexical verbs developing into auxiliaries, including negative and habitual ones, from a wide range of languages. The change from a nominalized constituent to a verbal one is documented in Gildea (2008) and Malchukov (2019). The confluence of these two changes are schematized in a general way in (93) and the schema in (94) illustrated the specific change we propose for Kalaallisut.

$$(93) [V-neq]_{nominal} V_{lexical} \Rightarrow [V-neq]_{verbal} Aux$$

$$(94) [V-neq]_{DP} ajor_{control} \Rightarrow [V-neq]_{TP} ajor_{aux}$$

We offer six observations in support of the hypothesized syntactic change in (94).

1. *Ajor-* as a control verb. Various dictionaries include a transitive entry for the verb *ajor-* along with an entry of its auxiliary usage. The Danish–Kalaallisut dictionary DAKA (<https://iserasuaat.gl/daka/>) lists the two entries in (95):²⁵

- (95) a. ajorpaa: klarer det ikke, magter det ikke
b. ajorpoq: *puijortarneq ajorpoq* ‘ryger ikke’

These entries are translated in (96).

- (96) a. ajorpaa: can't do it, doesn't have the capacity to
b. ajorpoq: *puiortarneq ajorpoq* 'does not smoke'

(96a) shows transitive agreement (-*paa* 3SG>3SG.IND), while (96b) shows intransitive agreement (-*poq* 3SG.IND). The meaning of transitive *ajor*- is similar to that of the control verbs *saper*- ‘can’t manage’ and *artor*- ‘be unable’. This, together with its transitive agreement, suggests that transitive *ajor*- in (95a) and (96a) is also a control verb. None of the four Kalallisut speakers we consulted recognize the transitive usage of *ajor*- in (96), including co-authors Author 3 and Author 4. We take this to suggest that for these speakers the grammaticalization schematized in (97) is completed and the original transitive control construction has been lost.

- (97) $[V\text{-}neg]_{DP} \text{ ajor}_{control} \Rightarrow [V\text{-}neg]_{TP} \text{ ajor}_{aux}$

2. Comparative evidence. The Comparative Eskimo Dictionary lists the Proto-Inuit root in (98) (Fortescue et al. 1994:60).

- (98) **ayuq-** ‘be unable to reach or do’²⁶

Cognates of this verb root with similar meanings are observed in four modern varieties of Inuit (Seward Peninsular Inuit, North Alaskan Inuit, Western Canadian Inuit, and Eastern Canadian Inuit) and possibly in the Alutiiq Alaskan variety of Yupik (Fortescue et al. 1994:60). However, aside from Kalaallisut, no modern Inuit variety employs the cognates of *ajor-* as a negative auxiliary (Richard Compton, p.c.). We take this to support our view that the PNC is an innovation in Kalallisut.

3. General function of *-neq* as a nominalizer. Further support for the PNC being an innovation comes from the fact that the overall distribution of *-neq* in Kalaallisut is as a nominalizer (Fortescue 1984: 44ff), making its verbal use in the PNC the odd one out. While we cannot give a full analysis of the nominalizing functions of *-neq*, we include the examples from Fortescue (1984) below to illustrate its range. Note that Fortescue glosses *-neq* with NOM for nominalizer. Also note that Fortescue uses the spelling *-niq/nir*, not the standard spelling *-neq/ner*.

- (99) piuma-sa-qar-**nir**-anit angi-niru-suq.
want-PASS.PART-have-NOM-3SG>3SG.ABL be.big-more-INTRANS.PART
bigger than he wanted Fortescue (1984: 103)

²⁵DAKA is an online dictionary comprising two dictionaries: *Oqaatsit* (1997) and *Ordbogen* (2003).

²⁶In the modern Kalaallisut orthography, this would be represented as *ajog-*, where the /q/ alternates with the /r/ when it precedes a suffix that begins with a consonant.

- (100) *atuarvi-qar-nir-mik inatsit*
 school-have-NOM-INST law
 the law pertaining to schools Fortescue (1984: 112)
- (101) *nipi-sia-ssa-tua-a-sara-aq timmia-nngu-it*
 sound-received-FUT-only-COP-HAB-3SG.IND bird-small-PL.ERG
qarlur-nir-at.
 twitter-NOM-3PL>3SG.ABS
 The only thing to be heard would be the twitter of little birds.
 Fortescue (1984: 183)

The fact that *-neq* functions as a nominalizer elsewhere supports our hypothesis that its nominalizing use in control constructions represents the original function of *-neq* and that its verbal use in the PNC is a localized innovation.

4. Grammaticalization of other negative control verbs. In descriptive works, the auxiliary *ajor-* is grouped with three other ‘negative verbs’ (Fortescue 1984:47–48, Kahn and Valijärvi 2022:245–247, and Sadock 2003:10–11, 33): *artor-* ‘to be unable’, *nalu-* ‘to not know’, and *saper-* ‘to not manage (to do something)’

Of these, *nalu-* and *saper-* can participate in PNC syntax, as observed in (102) and (103).

- (102) *Naja mersor-neq saper-poq.*
Naja.ABS sew-NEQ not.manage-3SG.IND
 ‘Naja can’t sew.’ Ellen Thrane CIT-1
- (103) *Naja biiler-neq nalu-voq.*
Naja.ABS drive-NEQ not.know-3SG.IND
 ‘Naja doesn’t know how to drive.’ Ellen Thrane BPA-2

These examples display PNC syntax in that the subject is in absolutive case (because the lexical verb is intransitive) and the agreement suffix of the negative verb is intransitive. In contrast, *artor-* only allows the syntax of control verbs, in which the subject has ergative case and the agreement suffix is transitive:

- (104) *Naja-p mersor-neq artor-paa.*
Naja-ERG sew-NEQ be.unable-3SG>3SG.IND
 ‘Naja is unable to sew.’ Ellen Thrane CKX-2
- (105) **Naja mersor-neq artor-poq.*
Naja.ABS sew-NEQ be.unable-3SG.IND
 Intended: ‘Naja is unable to sew.’ Ellen Thrane CIT-2

We can make sense of this pattern if the four negative verbs (*ajor-*, *artor-*, *nalu-*, *saper-*) originated as control verbs (cf. Fortescue et al. 1994:212) and three of them (*ajor-*, *nalu-*, *saper-*) evolved to become negative auxiliaries. In other words, there is a more general grammaticalization pathway from control verb to auxiliary in Kalaallisut.

5. Variable control-PNC syntax. Whereas *ajor-* has lost its meaning as a control verb and thus its ability to participate in control verb syntax, the negative auxiliaries (*nalu-* and *saper-*) preserve control verb syntax (106a–107a) but may alternatively participate in PNC syntax (106b–107b).

- (106) a. Naja-p biiler-neq nalu-vaa.
 Naja-ERG drive-NEQ not.know-3SG>3SG.IND
 ‘Naja doesn’t know how to drive.’ Ellen Thrane CIX-3
- b. Naja biiler-neq nalu-voq.
 Naja.ABS drive-NEQ not.know-3SG.IND
 ‘Naja doesn’t know how to drive.’ Ellen Thrane BPA-2
- (107) a. Naja-p mersor-neq saper-paa.
 Naja-ERG sew-NEQ not.manage-3SG>3SG.IND
 ‘Naja can’t sew.’ Ellen Thrane CLP-4
- b. Naja mersor-neq saper-poq.
 Naja.ABS sew-NEQ not.manage-3SG.IND
 ‘Naja can’t sew.’ Ellen Thrane CIT-1

The stages of development along the path to negative auxiliaries (PNC syntax) are indicated for each verb in Table 1.

	CONTROL	PNC
<i>artor-</i> ‘be unable’	✓	✗
<i>saper-</i> ‘not manage’	✓	✓
<i>nalu-</i> ‘no know’	✓	✓
<i>ajor-</i> NEG.HAB	✗	✓

Table 1: Degree of grammaticalization of the negative verbs

Following our proposal, the status of the verbs is understood as follows. One verb, *artor-*, has not participated in the change from control verb to auxiliary verb. Two verbs, *nalu-* and *saper-*, have begun the change, and one verb, *ajor-*, has completed the change.

6. Internal structure of *neq*-phrase. As established in section 6, PNC and control constructions exhibit identical restrictions on syntactic structure inside the *neq*-phrase. Under the hypothesized change from a negative control construction to the PNC, this follows from the Inertia Principle, which states that “syntactic change should not arise, unless it can be shown to be *caused*.” (Longobardi 2001:283; see also Keenan 2002, and Roberts 2017). The differences in case and agreement between control constructions and the PNC are examples of caused changes. Case and agreement depend directly on syntactic category: nominals require case and control agreement, verbs do not occur with case and do not control agreement. Thus, when a syntactic phrase changes category from a nominal element to a verbal element, this causes changes to the distribution of case and agreement. The differences in case and agreement documented in section 5 are the synchronic consequences of this change. On the other hand, nothing forces the amount of structure possible below *neq-* to change due to the category change. Therefore, by the Inertia Principle, it does not

change. Based on this reasoning, we claim that the fact that the restrictions on the structure internal to the *-neq* phrase is exactly the same in control constructions follows from the proposed change and therefore counts as an argument in its favor.

Semantic change. In the previous section we referenced the Inertia Principle, which states that syntactic change must be caused. On our proposal, that immediately raises a question: what caused the *-neq* phrase to change from a nominal element to a verbal element and, relatedly, what caused *ajor-* to change from a verb to an auxiliary? While we cannot offer proof, we would like to suggest that it was a semantic change that caused these syntactic changes. In particular, a semantic change from a modal negative meaning of *ajor-* ‘be unable to reach or do’ to just the negative meaning ‘not do’. This would be an instance of semantic bleaching, “whereby a more complex meaning is reduced to its nucleus” (Heine et al. 2019:295). This semantic change, we hypothesize, caused a syntactic category change of *ajor-* from V to Aux, which in turn caused a syntactic change in the syntactic status of the *-neq* from a nominal to verbal, as auxiliaries take verbal complements not nominal ones. This is admittedly speculative, but does, we think, add up to a coherent hypothesis about the diachrony of the PNC.

8 Conclusion and open issues

In this paper we have described and analyzed a periphrastic negation construction in Kalaallisut by way of a systematic comparison with a superficially similar control construction. Both constructions involve an embedded non-finite verb combined with the suffix *-neq*, but we have argued that the syntactic category of this suffix differs in the two constructions. In control constructions it is a nominalizing D head; in the PNC it is a non-finite T head. We have shown that this one difference accounts for a host of differences between the two constructions as regards case, agreement, word order, possessive morphology and subcategorization. The second part of the PNC is a finite negative auxiliary *ajor-* which we analyzed as a Neg head situated between T and C. This analysis explains why *ajor-* does not assign a theta-role (it is not a lexical verb); why it does not inflect for tense (it is above tense); why it does not co-occur with suffixal negation (they two morphemes occupy the same position).

In addition to this synchronic analysis we sketched a diachronic analysis according to which the PNC developed from a control structure by two interlocking processes: the *neq*-nominalization was reanalyzed as a non-finite TP and the negative control verb *ajor-* ‘can’t manage’ was grammaticalized into an auxiliary. In support of this analysis, we showed that other verbs designated ‘negative verbs’ in the descriptive and pedagogical literature are at various stages of the same change, though they all retain their control usage. Interestingly, there is evidence that some non-negative control verbs might be heading down the same path. Consider the example in (108)

- (108) Iipili-t neri-neq qatsup-pakka.
 apples-PL.ABS eat-NEQ be.tired.of-1SG>3PL.IND
 I am tired of eating apples.

Grethe Schmidt CZO-GS-5

Here the finite verb agrees with the third person plural absolutive internal argument of the embedded verb (*iipilit* ‘apples’), not with the *neq*-phrase. This is different from the regular control

construction where the control verb shows third person singular object agreement with its nominalized complement.

- (109) *Iipili*–tor–neq *qatsup*–**para**.
 apple–consume–NEQ be.tired.of–1SG>3SG.IND
 ‘I am tired of eating apples.’ Grethe Schmidt CIE-GS-6

In fact, (108) is parallel to the PNC example in (110), where the negative auxiliary shows third person plural object agreement with the internal argument of ‘eat’.

- (110) *Iipili*–t *neri*–neq *ajor*–**pakka**.
 apple–PL.ABS eat–NEQ HAB.NEG–1SG>3PL.IND
 I don’t eat apples. Grethe Schmidt DAM-GS-1

We tentatively suggest that the construction in (108) also involves reanalysis of the *neq*-phrase as a verbal projection (a non-finite TP) and restructuring of a biclausal control construction into a monoclausal structure. More work is needed to substantiate this analysis, but we can point to one piece of evidence in its favor, namely that it accounts for the contrast between (108) and (111).

- (111) **Iipili*–t *neri*–neq *qatsup*–para.
 apple–PL.ABS eat–NEQ be.tired.of–1SG>3SG.IND
 Intended: ‘I am tired of eating apples.’ Grethe Schmidt CIE-GS-4

In (111), *qatsup*- ‘be tired of’ shows third person singular object agreement. As argued at length in section 5.1, this is object agreement with the *neq*-phrase. In that structure an absolutive internal argument cannot be case licensed, because the unique absolutive case is assigned to the *neq*-phrase. However, in (108) an internal absolutive argument can be case licensed because the *neq*-phrase is a verbal projection and not in need of case.

A second avenue of future research is to investigate how our analysis of *-neq* in Kalaallisut relates to Carrier and Compton’s (2024) analysis of absolutive case assignment in nominalization structures formed with the Inuktitut cognate suffix *-niq*.

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